

Problem F. Double Strings

Time Limit 2000 ms

Mem Limit 262144 kB

You are given n strings $s_1, s_2, \dots, s_n$ of length at most 8.

For each string s_i , determine if there exist two strings s_j and s_k such that $s_i = s_j + s_k$. That is, s_i is the concatenation of s_j and s_k . Note that j **can** be equal to k .

Recall that the concatenation of strings s and t is $s + t = s_1s_2 \dots s_pt_1t_2 \dots t_q$, where p and q are the lengths of strings s and t respectively. For example, concatenation of "code" and "forces" is "codeforces".

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 10^5$) — the number of strings.

Then n lines follow, the i -th of which contains non-empty string s_i of length at most 8, consisting of lowercase English letters. Among the given n strings, there may be equal (duplicates).

The sum of n over all test cases doesn't exceed 10^5 .

Output

For each test case, output a binary string of length n . The i -th bit should be 1 if there exist two strings s_j and s_k where $s_i = s_j + s_k$, and 0 otherwise. Note that j can be equal to k .

Examples

Input	Output
3 5 abab ab abc abacb c 3 x xx xxx 8 codeforc es codes cod forc forces e code	10100 011 10100101

Note

In the first test case, we have the following:

- $s_1 = s_2 + s_2$, since **abab** = **ab** + **ab**. Remember that j can be equal to k .
- s_2 is not the concatenation of any two strings in the list.
- $s_3 = s_2 + s_5$, since **abc** = **ab** + **c**.
- s_4 is not the concatenation of any two strings in the list.
- s_5 is not the concatenation of any two strings in the list.

Since only s_1 and s_3 satisfy the conditions, only the first and third bits in the answer should be 1, so the answer is 10100.